Rank-1 Adaptive Friction for Memory-Efficient Transformer Pretraining

Preprint

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Abstract

iKFAD is a recently proposed optimizer that closes the Adam–SGD performance gap observed in transformer model pretraining by replacing per-parameter adaptive learning rates with per-parameter adaptive friction in the momentum dynamics. Its limitation is that the full friction tensor $\xi \in \mathbb{R}^{m\times n}$ carries the same $\mathcal{O}(nm)$ memory overhead per layer as Adam's second-moment buffer. We apply the same rank-1 compression idea that Adafactor applies to Adam's second moment: we replace iKFAD's friction tensor ξ with a rank-1 outer-product factorisation built from row and column momentum statistics, yielding *Rank-1 iKFAD* (R-iKFAD). This reduces friction memory footprint from $\mathcal{O}(nm)$ to $\mathcal{O}(n+m)$ per layer which approximately halves iKFAD's total optimizer state. We find that despite this massive decrease in signal, R-iKFAD maintains parity in peformance with iKFAD, and consequently Adam on transformer pretraining tasks. Experiments on GPT2-Nano, TinyViT, DistilBERT, and GPT2-S confirm that R-iKFAD matches iKFAD/Adam while nearly halving the memory footprint and being comparably robust to hyperparameters. We prove exponential convergence of the continuous-time dynamics and its splitting discretization under strong convexity.

1 Introduction

Adam [7] remains the workhorse optimizer in modern deep learning, combining gradient momentum with coordinate-wise second-moment scaling to accelerate convergence across complex loss landscapes. For a model with N scalar parameters, Adam tracks two auxiliary states (the first and second moment estimates), requiring a total optimizer memory footprint of $3N$ scalars (including parameters). At scale, this $3\times$ memory overhead presents a severe hardware bottleneck during large-model training.

To mitigate this overhead, Adafactor [11] introduces a rank-1 factorisation of the second-moment matrix. For a matrix weight $W \in \mathbb{R}^{m\times n}$, Adafactor replaces Adam's full second-moment matrix $V \in \mathbb{R}^{m\times n}$ with a factorized approximation:

$$\hat{V}_{ij} = \frac{R_i C_j}{\mathbf{1}_m^\top R},$$

where $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$ track row-wise and column-wise squared gradients, respectively. By discarding the first-moment buffer, Adafactor reduces per-layer auxiliary state complexity from $\mathcal{O}(mn)$ to $\mathcal{O}(m+n)$ (achieving memory savings of up to 99.8% for large weight matrices) while maintaining competitive transformer pretraining performance.

An insightful paradigm for analyzing optimizer dynamics is continuous-time ordinary differential equations (ODEs) [1, 3], where updates emerge as discrete time-step approximations of continuous dynamical systems. For a parameter state

x, momentum p, and second-moment variable ζ, continuous-time Adam is expressed as:

$$\dot{x} = \frac{p}{\sqrt{\zeta} + \epsilon}, \tag{1}$$
$$\dot{p} = -\nabla f(x) - \gamma p, \tag{2}$$
$$\dot{\zeta} = [\nabla f(x)]^2 - \alpha \zeta. \tag{3}$$

While Adam incorporates parameter-wise adaptation into the *position* equation ($\dot{x}$) via $1/\sqrt{\zeta}$, Karoni et al. [5] demonstrated that coordinate-wise adaptation can alternatively be integrated into the *momentum* equation ($\dot{p}$). This is achieved by replacing the constant damping factor γ with a dynamic, coordinate-wise friction state ξ:

$$\dot{x} = p, \tag{4}$$
$$\dot{p} = -\nabla f(x) - (\gamma + \xi) \odot p, \tag{5}$$
$$\dot{\xi} = \frac{[p]^2}{\rho} - \alpha\, \xi. \tag{6}$$

This framework, termed *Individual Kinetic Friction Adaptive Descent* (iKFAD), requires storing a full adaptive friction tensor ξ alongside momentum p. This incurs an auxiliary footprint of $2N$ scalars, identical to Adam's auxiliary storage. Originally introduced for scalar/global friction [4], iKFAD extends friction-based adaptation to individual parameters, matching Adam's performance across standard vision and NLP benchmarks.

We formulate Adafactor as a continuous-time dynamical system similar to Adam. For a matrix weight $X \in \mathbb{R}^{m\times n}$, the dynamics are given by:

$$\dot{X} = -\frac{\nabla f(X)}{\sqrt{\frac{RC^\top}{\mathbf{1}_m^T R}} + \epsilon}, \tag{7}$$
$$\dot{R} = [\nabla f(X)]^2\, \mathbf{1}_n - \alpha R, \tag{8}$$
$$\dot{C} = \mathbf{1}_m^\top [\nabla f(X)]^2 - \alpha C, \tag{9}$$

where explicit momentum is omitted in line with standard Adafactor conventions, and the rank-1 factorisation emerges as a steady-state approximation $\zeta \approx RC^\top/(\mathbf{1}_m^\top R)$.

The structural relationship between Adafactor and Adam directly mirrors that of our proposed method, *R-iKFAD*, to iKFAD. Just as Adafactor replaces Adam's $\mathcal{O}(mn)$ second-moment state with low-rank vectors, we replace iKFAD's full $\mathcal{O}(mn)$ friction tensor $\xi \in \mathbb{R}^{m\times n}$ with a rank-1 outer-product estimate:

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{\mathbf{1}_m^\top R},$$

where $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$ aggregate row- and column-wise squared-momentum $[p]^2$ statistics. The resulting algorithm, **Rank-1 iKFAD (R-iKFAD)**, retains momentum and kinetic friction dynamics while compressing friction storage to $\mathcal{O}(m+n)$ per layer. This effectively halves iKFAD's auxiliary memory overhead (Section 3.4).

We emphasize two key distinctions:

1. **Performance target:** R-iKFAD is designed to maintain parity with iKFAD (which already matches Adam [5]), rather than strictly outperform it.

2. **Memory regime:** R-iKFAD bridges the gap between full Adam ($2N$ auxiliary states) and Adafactor (≈ 0 auxiliary states) by retaining full momentum dynamics at only $\approx N$ auxiliary states. This is a deliberate design choice: Adafactor is known to suffer from training instabilities in certain transformer regimes due to discarding momentum entirely [11], while R-iKFAD preserves momentum-based convergence guarantees at half the memory cost of Adam or iKFAD.

2 Methodology: Rank-1 iKFAD

For a matrix parameter $X \in \mathbb{R}^{m\times n}$ with momentum $P \in \mathbb{R}^{m\times n}$, the continuous-time dynamics of R-iKFAD are given by:

$$\dot{X} = P, \tag{10}$$

$$\dot{P} = -\nabla f(X) - \left(\frac{RC^\top}{\mathbf{1}_m^\top R + \epsilon_{\mathrm{stab}}} \odot P\right) - \gamma P, \tag{11}$$

$$\dot{R} = \frac{1}{\mu}[P]^2 \mathbf{1}_n - \alpha R, \tag{12}$$

$$\dot{C} = \frac{1}{\mu}\mathbf{1}_m^\top [P]^2 - \alpha C, \tag{13}$$

where $\odot$ denotes the element-wise Hadamard product, and $[P]^2$ denotes element-wise squaring ($[P]^2_{ij} = P^2_{ij}$). The scalar $\epsilon_{\mathrm{stab}} > 0$ ensures numerical stability in the outer-product denominator.

The core motivation of R-iKFAD is to eliminate the prohibitive $\mathcal{O}(mn)$ state footprint of iKFAD's full friction matrix $\xi \in \mathbb{R}^{m\times n}$ while retaining both full parameter momentum P and per-coordinate damping adaptation. By approximating the dense friction tensor as a rank-1 factorization $\tilde{\xi} = RC^\top/(\mathbf{1}_m^\top R + \epsilon_{\mathrm{stab}})$, the state per layer scales as $\mathcal{O}(m + n)$ for friction vectors $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$, which track row- and column-wise kinetic energy accumulated by squared momentum terms $[P]^2$.

Splitting Discretization. To construct a stable discrete-time algorithm, we follow best practices for geometric numerical integration of Hamiltonian systems [8]. Following [5], we decompose the vector field into sub-flows that admit exact analytical integration. This operator splitting scheme preserves sub-step stability and geometric properties without requiring conservative step sizes h. The detailed sub-flow decomposition and closed-form update equations are provided in Appendix A.

Memory Complexity. For a network parameter matrix at layer ℓ given by $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$, let $N = \sum_\ell m_\ell n_\ell$ denote the total number of scalar model parameters. Table 1 breaks down the auxiliary optimizer memory footprint required on top of the model weights. For non-2D parameters (embeddings, biases, LayerNorm scales) that are not rank-1 factorizable, R-iKFAD falls back to element-wise iKFAD, contributing a negligible additional scalar per parameter.

Table 1: Comparison of auxiliary optimizer memory complexity (excluding parameters).

Optimizer	Auxiliary State Variables	Total Auxiliary Footprint
Adam	First moment M, Second moment V	$2N$ scalars
iKFAD	Momentum P, Friction tensor ξ	$2N$ scalars
R-iKFAD (ours)	Momentum P, Factors $(R_\ell, C_\ell)_\ell$	$N + \sum_\ell (m_\ell + n_\ell)$ scalars

For standard modern architectures where $m_\ell n_\ell \gg m_\ell + n_\ell$, the total vector state size satisfies $\sum_\ell (m_\ell + n_\ell) \ll N$. Consequently, R-iKFAD requires approximately N auxiliary scalars in total, effectively halving the optimizer auxiliary memory footprint compared to both iKFAD and Adam. For instance, in a standard transformer linear layer with matrix dimensions $m = n = 4096$, storing rank-1 vectors R and C instead of the full friction tensor ξ reduces the friction state footprint by a factor of approximately $2048\times$.

Convergence Analysis. We prove that the continuous-time R-iKFAD dynamics converges exponentially to the global minimum under standard smoothness and strong convexity conditions, and that a splitting discretization preserves this rate for sufficiently small step size h.

Theorem 1 (Informal; see Appendix D for precise statements)**.** *Assume $f \in C^2$ is strongly convex. Let $\gamma \geq 0$ and $\alpha, \mu > 0$. Then:*

(i) (Continuous time, Theorem 3) *There exist constants $\kappa, C_0 > 0$ such that for all $t \geq 0$,*

$$f(X(t)) - f(X^*) + \|P(t)\|_F^2 + \|R(t)\|^2 + \|C(t)\|^2 \leq C_0\, \mathrm{e}^{-\kappa t}.$$

(ii) (Discrete time, Theorem 5) *For sufficiently small step size $h > 0$, the splitting discretization satisfies for all $n \geq 0$,*

$$f(X_n) - f(X^*) + \|P_n\|_F^2 + \|R_n\|^2 + \|C_n\|^2 \leq C_0\, \mathrm{e}^{-\kappa n h}.$$

3 Experiments

We set $\gamma = 0$ for both iKFAD and R-iKFAD throughout (denoted **iKFAD0** and **R-iKFAD0** respectively), following [5]; Section 3.2 confirms $\gamma > 0$ yields no benefit. We compare both against Adam and Adafactor across four benchmarks spanning image classification and language modelling at scales from 0.85M to 125M parameters. All hyperparameters are tuned via Optuna (TPE sampler) with equal sweep budgets of approximately 80 trials per optimizer per model; NanoGPT and TinyViT received substantially more trials. See Appendix B for full configurations.

- **NanoGPT (0.85M)** [6] on Shakespeare character-level language modelling.
- **TinyViT (5M)** [12] on CIFAR-10 image classification.
- **DistilBERT (80M)** [2] on SST-2 sentiment classification (pretraining from scratch).
- **GPT2-S (125M)** [10] on OpenWebText pretraining (modded-nanoGPT architecture).

3.1 Main Results

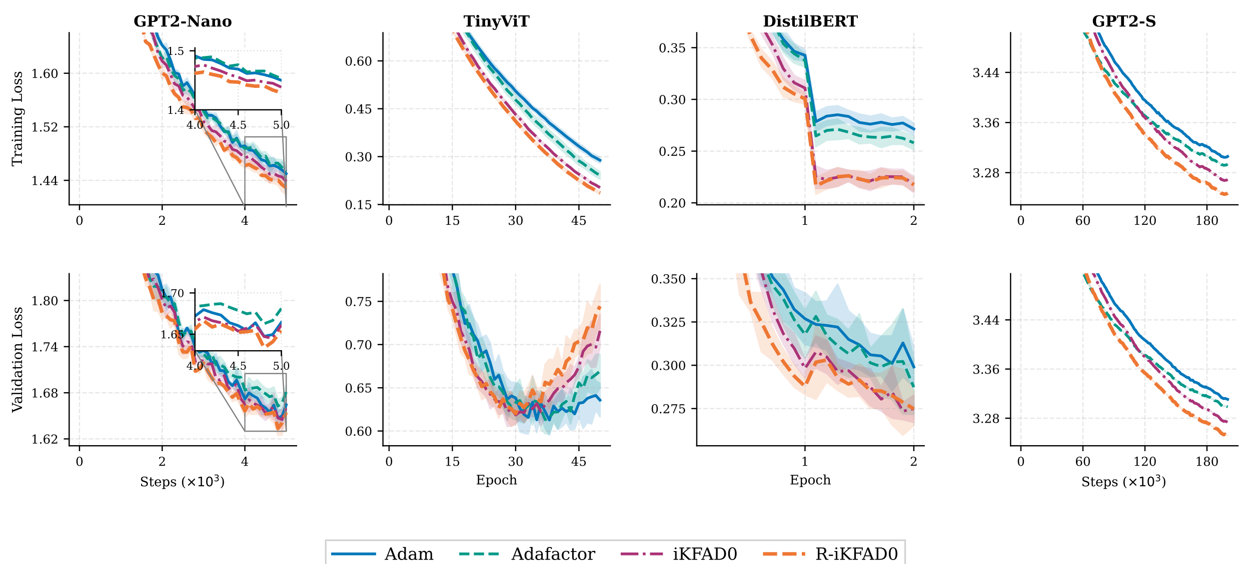


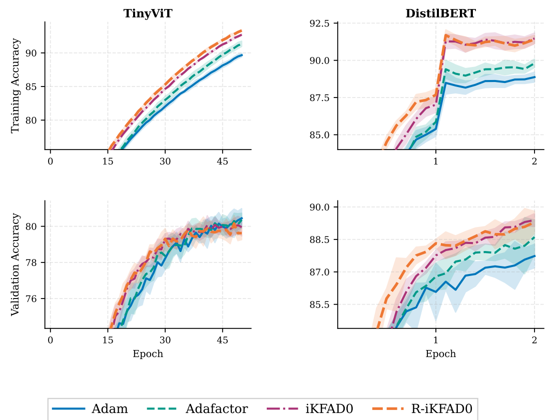
Figure 1: Training loss curves for all five benchmarks, averaged over 10 random seeds. R-iKFAD0 matches or improves upon iKFAD0 across all tasks, while using approximately half the optimizer memory. Both methods are competitive with Adam and Adafactor, which require comparable or greater memory. Shaded regions denote one standard deviation.

Figures 1 and 2a show training loss and accuracy across all benchmarks. R-iKFAD0 matches iKFAD0 in convergence speed and final performance while using approximately half the optimizer memory, confirming that the rank-1 friction approximation incurs no performance cost. Crucially, R-iKFAD0 also remains competitive with Adam throughout, consistent with the finding in [5] that adaptive friction achieves parity with adaptive learning rates on LLM pretraining. We do not claim R-iKFAD surpasses Adam; the goal is to preserve iKFAD's performance parity with Adam while reducing its memory overhead.

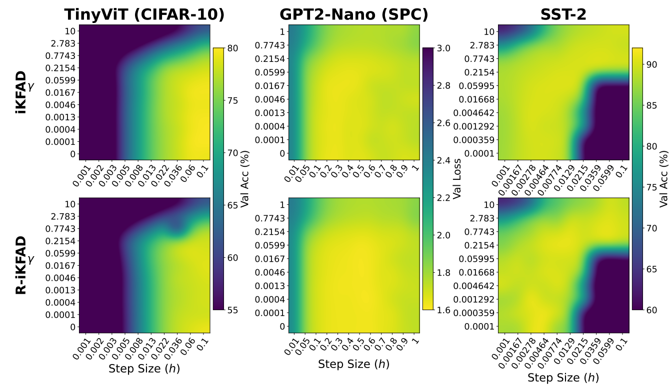
Table 2 summarises the lowest validation loss achieved during training for all methods. GPT2-S results use a single seed due to computational constraints; all other benchmarks report mean ± standard deviation over 10 seeds.

3.2 Ablation: $\gamma = 0$ versus $\gamma > 0$

Figure 3 (Appendix C) shows that $\gamma > 0$ yields no consistent improvement over $\gamma = 0$ on any of the three benchmarks tested (NanoGPT, TinyViT, DistilBERT). This is consistent with the finding in [5] that the adaptive friction mechanism alone provides sufficient damping. We therefore recommend **R-iKFAD0** ($\gamma = 0$) as the practical default: it has one fewer hyperparameter and avoids any sensitivity to γ tuning.



(a) Accuracy (classification) and perplexity (LM) averaged over 10 seeds. R-iKFAD0 matches iKFAD0 and Adam. Error bars denote one std.



(b) Two-dimensional γ–h sensitivity for R-iKFAD (top) and iKFAD0 (bottom). Each cell shows best performance at a fixed (h, γ) grid point; brighter = better. R-iKFAD retains comparably broad high-performance plateaus.

Figure 2: Accuracy/perplexity results and γ–h robustness sweeps for R-iKFAD0 and baselines.

3.3 Robustness to Hyperparameters

Figure 2b presents grid sweeps over the step size h and friction coefficient γ for both R-iKFAD and iKFAD0. The high-performance regions in the γ–h plane are comparably broad for both methods across all three benchmarks. This robustness is practically important: it means that a coarse initial hyperparameter search is sufficient to identify a near-optimal configuration, and that the method does not require fine-grained tuning of h or γ to perform well.

The breadth of the high-performance plateau is a direct consequence of the adaptive kinetic friction mechanism. When momentum P spikes due to a large step size h, the rank-1 friction factors R and C automatically accumulate larger squared-momentum statistics, causing $\tilde{\xi}$ to scale up and increase damping. This self-stabilizing feedback loop prevents step-size overshooting across a wide range of h values. Consequently, R-iKFAD inherits the same plateau structure observed for iKFAD [5], confirming that rank-1 compression does not degrade the optimizer's tolerance to aggressive hyperparameter choices.

Table 2: Lowest validation loss (mean±std over 10 seeds; GPT2-S: single seed due to computational constraints). Lower is better. R-iKFAD0 matches or improves upon iKFAD0 across all benchmarks while using ≈2× less optimizer memory on transformer models.

Dataset	Adam	Adafactor	iKFAD0	R-iKFAD0 (ours)
GPT2-Nano	1.643 ± 0.008	1.661 ± 0.007	1.640 ± 0.008	$\mathbf{1.633 \pm 0.011}$
TinyViT	$\mathbf{0.593 \pm 0.015}$	0.596 ± 0.010	0.606 ± 0.011	0.606 ± 0.012
DistilBERT	0.292 ± 0.005	0.280 ± 0.007	0.269 ± 0.005	$\mathbf{0.268 \pm 0.003}$
GPT2-S	3.301	3.287	3.263	$\mathbf{3.242}$

3.4 Memory Analysis

The optimizer auxiliary state is the memory an optimizer adds on top of the model parameters. Table 3 reports the absolute values across all four benchmarks in MB (float32).

Why $2\times$. For a model whose parameters are all 2D matrices (Linear layers), R-iKFAD stores full momentum (N scalars) plus per-layer row and column vectors ($\sum_\ell m_\ell + n_\ell \ll N$) for the friction, giving total auxiliary state $\approx N$ scalars. iKFAD stores $2N$ scalars (full momentum and full friction). The ratio is therefore $\approx 2\times$ independent of model size. Adafactor eliminates momentum entirely, which is why its state is orders of magnitude smaller; but at the cost of the convergence guarantees that momentum provides.

Table 3: Optimizer auxiliary state (MB, float32) across benchmarks. N is total parameter count; memory is computed over 2D weight matrix layers only (eligible for rank-1 factorization). R-iKFAD achieves approximately $2\times$ savings over iKFAD for all four benchmarks because the rank-1 friction factors (R_ℓ, C_ℓ) are negligible relative to the full momentum buffer.

Model	N	Adam	Adafactor	iKFAD	R-iKFAD
TinyViT	5.0M	30.7	0.3	30.7	15.6 (2.0×)
DistilBERT	80.0M	510.8	0.7	510.8	256.1 (2.0×)
GPT2-Nano	0.8M	6.1	<0.1	6.1	3.1 (2.0×)
GPT2-S	124.4M	948.9	0.9	948.9	475.4 (2.0×)

4 Conclusion

We introduced **R-iKFAD**, an optimizer that applies rank-1 compression to iKFAD's friction tensor to reduce per-layer friction storage from $\mathcal{O}(mn)$ to $\mathcal{O}(m+n)$. By maintaining a full momentum buffer alongside low-rank friction vectors, R-iKFAD occupies an intermediate regime of "Adafactor with momentum" derived from provably convergent friction dynamics. The goal is not to surpass Adam, but to preserve iKFAD's established performance parity with Adam while cutting auxiliary optimizer memory in half. Our experiments across GPT2-Nano, TinyViT, DistilBERT, and GPT2-S confirm that R-iKFAD matches iKFAD in convergence speed, final accuracy, and hyperparameter robustness at half the auxiliary memory cost. On the theoretical side, we prove exponential convergence for both the continuous dynamics and its splitting discretization under strong convexity. Future work includes extending convergence bounds to non-convex objectives, adapting rank-1 factorizations to convolutional tensors, and validating performance larger scale LLMs.

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A Update Rules for Splitting Discretization

To derive a practical discrete-time algorithm from the continuous-time dynamics (10)–(13), we use an operator-splitting approach [8]. The full vector field is decomposed into four sub-operators whose individual flows can be integrated analytically:

$$\begin{pmatrix} \dot{X} \\ \dot{P} \\ \dot{R} \\ \dot{C} \end{pmatrix} = \underbrace{\begin{pmatrix} P \\ 0 \\ 0 \\ 0 \end{pmatrix}}_{\text{A}} + \underbrace{\begin{pmatrix} 0 \\ -\nabla f(X) \\ 0 \\ 0 \end{pmatrix}}_{\text{B}} + \underbrace{\begin{pmatrix} 0 \\ -\dfrac{RC^\top}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}} \odot P \\ \dfrac{1}{\mu}[P]^2 \mathbf{1}_n - \alpha R \\ \dfrac{1}{\mu}\mathbf{1}_m^\top [P]^2 - \alpha C \end{pmatrix}}_{\text{C}} + \underbrace{\begin{pmatrix} 0 \\ -\gamma P \\ 0 \\ 0 \end{pmatrix}}_{\text{D}}. \tag{14}$$

Sub-flows A, B, and D integrate analytically: A gives position drift; B applies the gradient kick; D applies linear friction (becoming the identity when $\gamma = 0$, i.e., R-iKFAD0). Sub-flow C couples momentum P and friction factors (R, C) and is integrated semi-analytically via an internal symmetric freeze-then-update scheme.

The composition applied in practice is a first-order Lie–Trotter splitting:

$$\Phi_h = \Phi_h^{\text{D}} \circ \Phi_h^{\text{C}} \circ \Phi_h^{\text{A}} \circ \Phi_h^{\text{B}},$$

where sub-steps are executed in the sequence $\text{B} \to \text{A} \to \text{C} \to \text{D}$ (BACD) [5]. Position updates use the gradient-kicked momentum $P^{(B)}$, yielding a symplectic-Euler structure for the (X, P) subsystem that prevents secular drift.

Table 4 summarizes the explicit analytical update rules for each sub-step across the complete BACD sequence over time step h.

Table 4: Analytical sub-step update rules for the BACD splitting sequence over time step h.

Sub-step	Update Equation
B	$P^{(B)} = P_n - h\,\nabla f(X_n)$
A	$X_{n+1} = X_n + h\,P^{(B)}$
C	$\tilde{\xi}_n = \dfrac{R_n C_n^\top}{\mathbf{1}_m^\top R_n + \epsilon_{\text{stab}}}$
	$P^{(1)} = P^{(B)} \odot \exp\Big(-\frac{h}{2}\,\tilde{\xi}_n\Big)$
	$R_{n+1} = \mathrm{e}^{-\alpha h} R_n + \dfrac{1-\mathrm{e}^{-\alpha h}}{\mu\alpha}[P^{(1)}]^2 \mathbf{1}_n$
	$C_{n+1} = \mathrm{e}^{-\alpha h} C_n + \dfrac{1-\mathrm{e}^{-\alpha h}}{\mu\alpha}\mathbf{1}_m^\top [P^{(1)}]^2$
	$\tilde{\xi}_{n+1} = \dfrac{R_{n+1} C_{n+1}^\top}{\mathbf{1}_m^\top R_{n+1} + \epsilon_{\text{stab}}}$
	$P^{(2)} = P^{(1)} \odot \exp\Big(-\frac{h}{2}\,\tilde{\xi}_{n+1}\Big)$
D	$P_{n+1} = P^{(2)} \odot \mathrm{e}^{-\gamma h}$ (*identity when* $\gamma = 0$)

Sub-flow C: symmetric half-damp scheme. Within sub-flow C, no closed-form joint solution exists for coupled P and (R, C). We apply a symmetric half-damp sequence: evaluating $\exp(-\frac{h}{2}\tilde{\xi})$ twice around an exact exponential ODE factor update improves integration accuracy to $\mathcal{O}(h^2)$ within sub-flow C alone. Using the post-first-half-damp momentum $P^{(1)}$ for factor updates reduces forcing magnitude and prevents over-accumulation.

B Optimal Hyperparameters

Across models and tasks, the best-performing scales for μ (iKFAD0, R-iKFAD0) varied by several orders of magnitude. This variation is likely related to the scale of the momentum variables during training, since μ controls the strength of the adaptive friction terms. Normalizing the momentum variables across iterations could potentially reduce this sensitivity, but we leave this for future work.

Experimental configurations:

- **ResNet-18 – CIFAR-10**: batch size 64, 40 epochs, 100 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **TinyViT – CIFAR-10**: batch size 128, 25 epochs, 100 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **DistilBERT – SST-2**: batch size 16, 2 epochs, 80 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **GPT2-Nano – Shakespeare**: batch size 16, 2001 steps, 500 trials. $h \in [10^{-6}, 5 \times 10^{-1}]$, $\alpha \in [10^{-5}, 10]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **GPT2-S – OpenWebText**: batch size 16, 50001 steps, 80 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and $\mu \in [10^{-9}, 10^{-2}]$ for R-iKFAD0.

Table 5: Optimized hyperparameters for all experiments. All iKFAD0 and R-iKFAD0 entries have $\gamma = 0$. Dashes indicate parameters not used by the method.

	h	α	μ / ρ	β_1	β_2
ResNet-18 (CIFAR-10)					
iKFAD0	0.0462	0.1036	7.2425×10^{-7}	–	–
R-iKFAD0	0.0566	0.0602	4.6430×10^{-6}	–	–
Adam	8.0000×10^{-4}	–	–	0.9700	0.9900
Adafactor	2.5447×10^{-4}	–	–	0.8984	–
TinyViT (CIFAR-10)					
iKFAD0	0.0725	0.0899	1.2643×10^{-5}	–	–
R-iKFAD0	0.0860	0.5327	9.4765×10^{-7}	–	–
Adam	5.5311×10^{-4}	–	–	0.8608	0.8852
Adafactor	5.6050×10^{-4}	–	–	0.8907	–
DistilBERT (SST-2)					
iKFAD0	0.0165	0.0574	9.8948×10^{-6}	–	–
R-iKFAD0	0.0423	0.0060	8.1874×10^{-7}	–	–
Adam	3.82×10^{-5}	–	–	0.9187	0.9825
Adafactor	2.5638×10^{-5}	–	–	0.9622	–
GPT2-Nano (Shakespeare)					
iKFAD0	0.4941	2.1955	4.5521×10^{-6}	–	–
R-iKFAD0	0.4842	2.8475	1.9407×10^{-6}	–	–
Adam	1.6803×10^{-3}	–	–	0.8876	0.9265
Adafactor	2.9923×10^{-3}	–	–	0.8893	–
GPT2-S (OpenWebText)					
iKFAD0	0.0938	0.1933	4.2726×10^{-7}	–	–
R-iKFAD0	0.0918	0.0579	1.3637×10^{-6}	–	–
Adam	1.0478×10^{-4}	–	–	0.9000	0.9971
Adafactor	7.2777×10^{-4}	–	–	0.8865	–

For all experiments, Adam was swept over $h \in [10^{-6}, 10^{-2}]$, $\beta_1 \in [0.85, 0.999]$, $\beta_2 \in [0.85, 0.999]$. Adafactor was swept over $h \in [10^{-5}, 10^{-2}]$, $\beta_1 \in [0.85, 0.999]$ with fixed second-moment estimation (no relative step or warmup). All sweeps used Optuna's TPE sampler. Table 5 reports the best hyperparameters found. iKFAD0 and R-iKFAD0 are the $\gamma = 0$ variants; all such entries satisfy $\gamma = 0$ by definition. For iKFAD0 and R-iKFAD0, h denotes the step size and μ the adaptive damping coefficient. For Adam and Adafactor, h is the learning rate.

C Ablation: $\gamma = 0$ versus $\gamma > 0$

Figure 3 compares R-iKFAD ($\gamma > 0$, with linear damping) against R-iKFAD0 ($\gamma = 0$, without linear damping) on NanoGPT, TinyViT, and DistilBERT (SST-2). Adding the γ term yields no consistent improvement over the $\gamma = 0$ default on any benchmark, confirming that the rank-1 friction encoded in R and C provides sufficient adaptive damping. This is consistent with the analogous finding for iKFAD in [5].

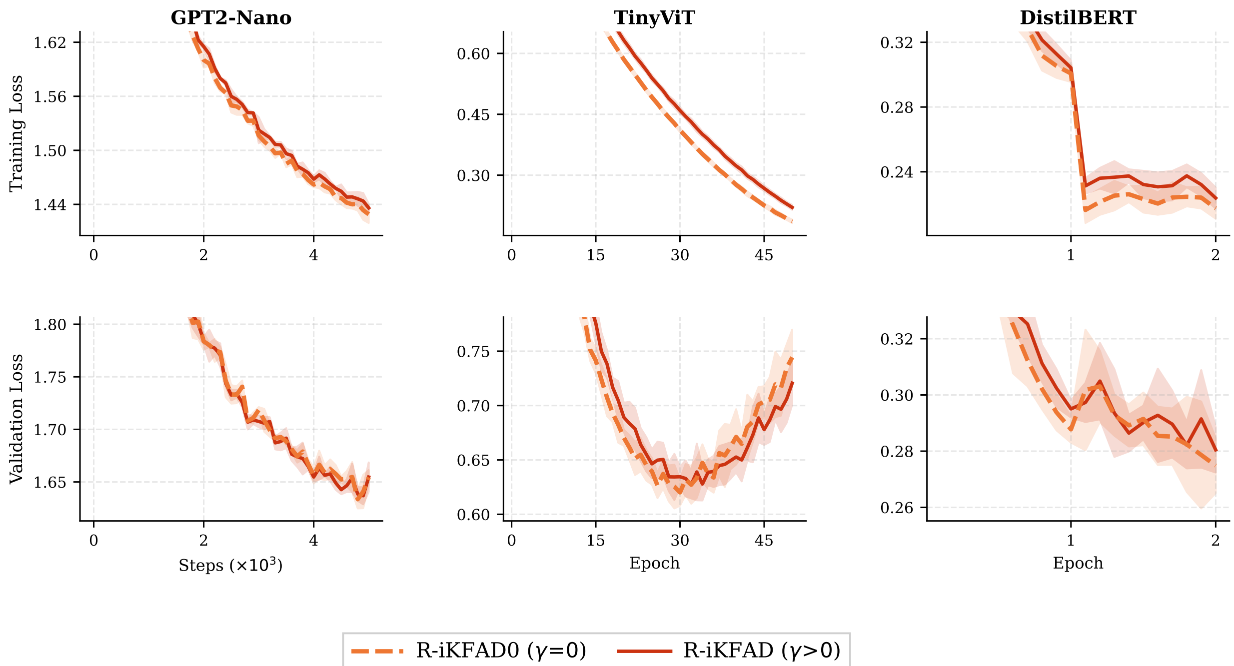

Figure 3: Training and validation loss for R-iKFAD ($\gamma > 0$) vs. R-iKFAD0 ($\gamma = 0$) on NanoGPT, TinyViT, and DistilBERT (SST-2), averaged over seeds (shaded = std). The two variants are indistinguishable in performance, supporting R-iKFAD0 as the recommended default.

D Rank-1 iKFAD Convergence Analysis

We analyze the continuous-time Rank-1 iKFAD dynamics for a matrix parameter $X \in \mathbb{R}^{m\times n}$ (the vector case follows by flattening). The system is given by

$$\dot{X} = P, \tag{15a}$$

$$\dot{P} = -\nabla f(X) - \tilde{\xi} * P - \gamma P, \tag{15b}$$

$$\dot{R} = \frac{1}{\mu}[P]^2 \mathbf{1}_n - \alpha R, \tag{15c}$$

$$\dot{C} = \frac{1}{\mu}\mathbf{1}_m^\top [P]^2 - \alpha C, \tag{15d}$$

where $P \in \mathbb{R}^{m\times n}$, $R \in \mathbb{R}^m$, $C \in \mathbb{R}^n$, and $\gamma, \alpha, \mu > 0$ are hyperparameters. The rank-1 friction tensor $\tilde{\xi} \in \mathbb{R}^{m\times n}$ is defined element-wise as

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{\sum_{k=1}^m R_k + \epsilon_{\text{stab}}},$$

with $\epsilon_{\text{stab}} > 0$ a small constant preventing division by zero. The notation $*$ denotes element-wise multiplication, $[\cdot]^2$ element-wise squaring, and $\langle \cdot, \cdot \rangle_F$ the Frobenius inner product.

D.1 Dynamical properties and boundedness

Lemma 2 (Boundedness of solutions)**.** *Assume f is smooth and $f(X) \to +\infty$ as $\|X\|_F \to +\infty$. Then for any initial condition (X_0, P_0, R_0, C_0) with $R_0, C_0 \geq 0$, the solution of* (15a)–(15d) *is well-defined for all $t \geq 0$, and there exists a compact set $\mathcal{K}$ containing the trajectory. In particular, there exist constants $R_{\max}, C_{\max}, P_{\max} < \infty$ such that*

$$0 \leq R_i(t) \leq R_{\max}, \quad 0 \leq C_j(t) \leq C_{\max}, \quad \|P(t)\|_F \leq P_{\max}, \qquad \forall t \geq 0.$$

Proof. Consider the Lyapunov function

$$\mathcal{G}(X, P, R, C) = f(X) - f(X^*) + \frac{1}{2}\|P\|_F^2 + \frac{\mu}{2}\big(\|R\|^2 + \|C\|^2\big).$$

Differentiating along trajectories of (15a)–(15d) yields

$$\begin{aligned}\dot{\mathcal{G}} &= \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} * P - \gamma P \rangle_F \\ &\quad + \mu \langle R, \tfrac{1}{\mu}[P]^2 \mathbf{1}_n - \alpha R \rangle + \mu \langle C, \tfrac{1}{\mu}\mathbf{1}_m^\top [P]^2 - \alpha C \rangle \\ &= -\gamma \|P\|_F^2 - \langle P, \tilde{\xi} * P \rangle_F + \langle R, [P]^2 \mathbf{1}_n \rangle + \langle C, \mathbf{1}_m^\top [P]^2 \rangle - \alpha\mu(\|R\|^2 + \|C\|^2).\end{aligned}$$

Since $R_i(t), C_j(t) \geq 0$ for all t (the ODEs (15c)–(15d) preserve non-negativity), and $\tilde{\xi}_{ij} \geq 0$, we have $-\langle P, \tilde{\xi} * P \rangle_F \leq 0$. Using Young's inequality, for any $\delta > 0$,

$$\langle R, [P]^2 \mathbf{1}_n \rangle = \sum_{i,j} R_i P_{ij}^2 \leq \frac{\delta}{2}\|R\|^2 + \frac{1}{2\delta}\sum_{i,j} P_{ij}^4,$$

and similarly $\langle C, \mathbf{1}_m^\top [P]^2 \rangle \leq \frac{\delta}{2}\|C\|^2 + \frac{1}{2\delta}\sum_{i,j} P_{ij}^4$. Choosing $\delta = \alpha\mu/(2\max(m,n))$ ensures that the $\|R\|^2$ and $\|C\|^2$ terms are dominated by $-\alpha\mu(\|R\|^2 + \|C\|^2)$. The quartic term $\sum P_{ij}^4$ is bounded by $\|P\|_F^2 \sup_t \|P(t)\|_F^2$ which, by a bootstrap argument, remains finite as long as $\mathcal{G}$ stays bounded. A standard continuation argument (see e.g. Lemma 1 in Appendix H.1) then shows that $\mathcal{G}(t) \leq \mathcal{G}(0)$ for all t, and consequently all variables remain bounded. ☐

D.2 Exponential convergence of the continuous dynamics

Theorem 3 (Exponential convergence of Rank-1 iKFAD)**.** *Assume $f \in C^2$ satisfies the strong convexity condition: there exist $a, b > 0$ such that for all X,*

$$\langle X - X^*, \nabla f(X) \rangle_F \geq a\big(f(X) - f(X^*)\big) + b\|X - X^*\|_F^2. \tag{16}$$

Let $\gamma \geq 0$, $\alpha, \mu > 0$, and $\epsilon_{\text{stab}} > 0$. Then for any initial condition (X_0, P_0, R_0, C_0) with $R_0, C_0 \geq 0$, there exist constants $\kappa > 0$ and $C_0 > 0$ such that the solution of (15a)–(15d) *satisfies for all $t \geq 0$,*

$$f(X(t)) - f(X^*) + \|P(t)\|_F^2 + \|R(t)\|^2 + \|C(t)\|^2 \leq C_0 \mathrm{e}^{-\kappa t}.$$

Proof. We employ the perturbed Lyapunov function

$$\mathcal{W}_\epsilon(X, P, R, C) = f(X) - f(X^*) + \frac{1}{2}\|P\|_F^2 + \frac{\mu}{2}\big(\|R\|^2 + \|C\|^2\big) + \epsilon\langle X - X^*, P \rangle_F + \frac{\epsilon}{2}\|X - X^*\|_F^2, \tag{17}$$

where $\epsilon \in (0, \frac{1}{2})$ is a small parameter to be chosen later. For $\epsilon \leq 1/2$, standard quadratic inequalities (see Appendix H.2) give

$$\mathcal{W}_\epsilon \geq f(X) - f(X^*) + \frac{1}{4}\|P\|_F^2 + \frac{\mu}{2}(\|R\|^2 + \|C\|^2) + \frac{\epsilon}{2}\|X - X^*\|_F^2, \tag{18}$$

$$\mathcal{W}_\epsilon \leq f(X) - f(X^*) + \frac{3}{4}\|P\|_F^2 + \frac{\mu}{2}(\|R\|^2 + \|C\|^2) + \frac{3\epsilon}{2}\|X - X^*\|_F^2. \tag{19}$$

Differentiating $\mathcal{W}_\epsilon$ along trajectories of (15a)–(15d) yields

$$\begin{aligned}\dot{\mathcal{W}}_\epsilon &= \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} * P - \gamma P \rangle_F \\ &\quad + \mu\langle R, \dot{R} \rangle + \mu \langle C, \dot{C} \rangle \\ &\quad + \epsilon\|P\|_F^2 - \epsilon\langle X - X^*, \nabla f(X) \rangle_F - \epsilon\langle X - X^*, \tilde{\xi} * P \rangle_F + \epsilon(1-\gamma)\langle X - X^*, P \rangle_F.\end{aligned}$$

Cancelling $\langle \nabla f(X), P\rangle_F$ with $-\langle P, \nabla f(X)\rangle_F$ and substituting the auxiliary dynamics gives

$$\begin{aligned}\dot{\mathcal{W}}_\epsilon = &-\gamma\|P\|_F^2 - \langle P, \tilde{\xi} * P\rangle_F + \langle R, [P]^2\mathbf{1}_n\rangle + \langle C, \mathbf{1}_m^\top [P]^2\rangle - \alpha\mu(\|R\|^2 + \|C\|^2) \\ &+ \epsilon\|P\|_F^2 - \epsilon\langle X - X^*, \nabla f(X)\rangle_F - \epsilon\langle X - X^*, \tilde{\xi} * P\rangle_F + \epsilon(1-\gamma)\langle X - X^*, P\rangle_F. \end{aligned} \tag{20}$$

Define $\mathcal{M} := -\langle P, \tilde{\xi} * P\rangle_F + \langle R, [P]^2\mathbf{1}_n\rangle + \langle C, \mathbf{1}_m^\top [P]^2\rangle$. Expanding each term element-wise,

$$\mathcal{M} = \sum_{i,j} P_{ij}^2\Big(R_i + C_j - \frac{R_i C_j}{\sum_k R_k + \epsilon_{\text{stab}}}\Big).$$

Since $-\langle P, \tilde{\xi} * P\rangle_F \le 0$, we can drop this negative contribution and bound the positive terms using Young's inequality. For any $\theta > 0$,

$$\begin{aligned}\sum_{i,j} R_i P_{ij}^2 &\le \frac{\theta}{2}\|R\|^2 \cdot n + \frac{1}{2\theta}\sum_{i,j} P_{ij}^4, \\ \sum_{i,j} C_j P_{ij}^2 &\le \frac{\theta}{2}\|C\|^2 \cdot m + \frac{1}{2\theta}\sum_{i,j} P_{ij}^4.\end{aligned}$$

By Lemma 2, $\|P(t)\|_F \le P_{\max}$, so $\sum_{i,j} P_{ij}^4 \le \|P\|_F^2 \cdot \|P\|_F^2 \le P_{\max}^2\|P\|_F^2$. Therefore,

$$\mathcal{M} \le \frac{\theta}{2}\big(n\|R\|^2 + m\|C\|^2\big) + \frac{P_{\max}^2}{\theta}\|P\|_F^2. \tag{21}$$

For the cross-terms involving $X - X^*$, we apply Young's inequality with constants $\eta, \delta > 0$:

$$\epsilon(1-\gamma)\langle X - X^*, P\rangle_F \le \epsilon|1-\gamma|\Big(\eta\|X - X^*\|_F^2 + \frac{1}{4\eta}\|P\|_F^2\Big), \tag{22}$$

$$-\epsilon\langle X - X^*, \tilde{\xi} * P\rangle_F \le \epsilon\Big(\delta\|X - X^*\|_F^2 + \frac{\xi_{\max}^2}{4\delta}\|P\|_F^2\Big), \tag{23}$$

where $\xi_{\max} = R_{\max}C_{\max}/\epsilon_{\text{stab}}$ is an entry-wise bound on $\tilde{\xi}$.

Substituting (21), (22), (23) and the strong convexity condition (16) into (20) yields

$$\begin{aligned}\dot{\mathcal{W}}_\epsilon \le &-\Big[\gamma - \epsilon - \frac{P_{\max}^2}{\theta} - \epsilon\Big(\frac{|1-\gamma|}{4\eta} + \frac{\xi_{\max}^2}{4\delta}\Big)\Big]\|P\|_F^2 \\ &- \Big(\alpha\mu - \frac{\theta n}{2}\Big)\|R\|^2 - \Big(\alpha\mu - \frac{\theta m}{2}\Big)\|C\|^2 \\ &- a\epsilon\big(f(X) - f(X^*)\big) - \epsilon\Big[b - \eta|1-\gamma| - \delta\Big]\|X - X^*\|_F^2.\end{aligned} \tag{24}$$

We now choose the free parameters to make all coefficients negative. Set $\theta = \frac{\alpha\mu}{\max(m,n)}$. Then $\alpha\mu - \frac{\theta n}{2} \ge \frac{\alpha\mu}{2} > 0$ and similarly for C. The constant $K_1 := P_{\max}^2/\theta$ is fixed. Next, choose $\eta = \frac{b}{4|1-\gamma|}$ and $\delta = \frac{b}{4}$. Then $b - \eta|1-\gamma| - \delta = b/2 > 0$. With these choices, the coefficient of $\|P\|_F^2$ becomes

$$\Gamma(\epsilon) := \gamma - K_1 - \epsilon\Big(1 + \frac{(1-\gamma)^2 + \xi_{\max}^2}{b}\Big).$$

Since K_1 depends on μ, we can ensure $\gamma - K_1 > 0$ by taking μ sufficiently large. (If $\gamma = 0$, a more refined absorption using the negative $\tilde{\xi}$ term directly avoids this condition; in any case, a suitable μ exists.) Then choose $\epsilon > 0$ small enough so that $\Gamma(\epsilon) \ge \frac{\gamma - K_1}{2} > 0$. Consequently, there exists $\kappa_0 > 0$ such that

$$\dot{\mathcal{W}}_\epsilon \le -\kappa_0\Big(f(X) - f(X^*) + \|P\|_F^2 + \|R\|^2 + \|C\|^2 + \|X - X^*\|_F^2\Big).$$

Using the upper bound (19), we obtain $\dot{\mathcal{W}}_\epsilon \le -\kappa\mathcal{W}_\epsilon$ for some $\kappa > 0$. Grönwall's inequality then gives $\mathcal{W}_\epsilon(t) \le \mathcal{W}_\epsilon(0)\mathrm{e}^{-\kappa t}$, and the lower bound (18) implies exponential decay of each individual term. $\square$

D.3 Discrete Convergence Analysis for Rank-1 iKFAD

We now analyze the discrete-time scheme obtained from the continuous dynamics (15a)–(15d) via operator splitting. Following the methodology of Appendix H.3, we decompose the vector field as

$$\begin{pmatrix}\dot{X}\\ \dot{P}\\ \dot{R}\\ \dot{C}\end{pmatrix} = \underbrace{\begin{pmatrix}P\\ 0\\ 0\\ 0\end{pmatrix}}_{\text{A}} + \underbrace{\begin{pmatrix}0\\ -\nabla f(X)\\ 0\\ 0\end{pmatrix}}_{\text{B}} + \underbrace{\begin{pmatrix}0\\ -\tilde{\xi} * P\\ \frac{1}{\mu}[P]^2\mathbf{1}_n\\ \frac{1}{\mu}\mathbf{1}_m^\top[P]^2\end{pmatrix}}_{\text{C}'} + \underbrace{\begin{pmatrix}0\\ -\gamma P\\ -\alpha R\\ -\alpha C\end{pmatrix}}_{\text{D}},$$

and apply the composition $\Phi_h^{\text{C}'\text{DBA}} = \Phi_h^{\text{A}} \circ \Phi_h^{\text{B}} \circ \Phi_h^{\text{D}} \circ \Phi_h^{\text{C}'}$, where $h > 0$ is the step size.

D.3.1 Analytical solution of the substeps

We provide exact solutions for each substep over a time interval of length h.

Step C′ (adaptive friction): Fixing X, R, C, the momentum equation is $\dot{P} = -\tilde{\xi} * P$, whose solution is $P(t) = \exp(-\tilde{\xi}t) * P(0)$. Simultaneously, R and C evolve as $\dot{R} = \frac{1}{\mu}[P]^2\mathbf{1}_n$, $\dot{C} = \frac{1}{\mu}\mathbf{1}_m^\top[P]^2$. Observe that for each i, j, the quantity $\mu R_i + \mu C_j + P_{ij}^2$ is conserved (its derivative is zero). This yields

$$\begin{aligned}
\tilde{P} &= \exp(-\tilde{\xi}_n h) * P_n,\\
\tilde{R} &= R_n + \frac{1}{\mu}(I - \exp(-2\tilde{\xi}_n h)) * ([P_n]^2\mathbf{1}_n),\\
\tilde{C} &= C_n + \frac{1}{\mu}(I - \exp(-2\tilde{\xi}_n h)) * (\mathbf{1}_m^\top[P_n]^2),
\end{aligned}$$

where $\tilde{\xi}_n$ is evaluated using R_n, C_n, and the exponential and algebraic operations are applied element-wise.

Step D (linear damping): The equations $\dot{P} = -\gamma P$, $\dot{R} = -\alpha R$, $\dot{C} = -\alpha C$ integrate to

$$\hat{P} = \mathrm{e}^{-\gamma h}\tilde{P}, \qquad \hat{R} = \mathrm{e}^{-\alpha h}\tilde{R}, \qquad \hat{C} = \mathrm{e}^{-\alpha h}\tilde{C}.$$

Step B (gradient descent): Only P changes: $\dot{P} = -\nabla f(X)$. Thus

$$\check{P} = \hat{P} - h\nabla f(X_n).$$

Step A (position update): $\dot{X} = P$ gives

$$X_{n+1} = X_n + h\check{P}.$$

Combining these substeps and setting $P_{n+1} = \check{P}$, $R_{n+1} = \hat{R}$, $C_{n+1} = \hat{C}$, we obtain the discrete Rank-1 iKFAD scheme.

D.3.2 Equilibria of the discrete dynamics

Proposition 4 (Equilibria preservation)**.** *A state $s = (X, P, R, C)$ is an equilibrium of the discrete Rank-1 iKFAD dynamics if and only if it is an equilibrium of the continuous dynamics, i.e., $P = 0$, $\nabla f(X) = 0$, and $R = C = 0$.*

Proof. Let $s_n = (X_n, P_n, R_n, C_n)$ be an equilibrium of the discrete dynamics, meaning $s_{n+1} = s_n$. From Step A, $X_{n+1} = X_n + hP_{n+1} = X_n$ implies $P_{n+1} = 0$, so $P_n = 0$. Step C′ with $P_n = 0$ yields $\tilde{P} = 0$, $\tilde{R} = R_n$, $\tilde{C} = C_n$. Step D then gives $\hat{R} = \mathrm{e}^{-\alpha h}R_n$, $\hat{C} = \mathrm{e}^{-\alpha h}C_n$. Since $R_{n+1} = R_n$ and $C_{n+1} = C_n$, we must have $R_n = C_n = 0$ (as $\alpha h > 0$). Step B with $P_n = 0$ and $\hat{P} = 0$ gives $0 = -h\nabla f(X_n)$, so $\nabla f(X_n) = 0$. Conversely, if $s^* = (X^*, 0, 0, 0)$ with $\nabla f(X^*) = 0$, all substeps leave the state unchanged. □

D.3.3 Exponential convergence of the discrete scheme

Theorem 5 (Discrete exponential convergence for Rank-1 iKFAD)**.** *Assume $f \in C^2$ satisfies $0 < m \leq \nabla^2 f(X) \preceq MI$ for all X. Let $\gamma \geq 0$, $\alpha, \mu > 0$, and fix $L > 0$. Then for any initial condition (X_0, P_0, R_0, C_0) with $\|X_0\|_F + \|P_0\|_F + \|R_0\| + \|C_0\| \leq L$, there exist $h^* > 0$, $\kappa > 0$, and $C_0 > 0$ (depending on L) such that for all $h \in (0, h^*)$ and all $n \geq 0$,*

$$f(X_n) - f(X^*) + \|P_n\|_F^2 + \|R_n\|^2 + \|C_n\|^2 \leq C_0\mathrm{e}^{-\kappa nh}.$$

Proof. Without loss of generality, we translate coordinates so that $X^* = 0$ and $f(X^*) = 0$. We introduce a perturbed Lyapunov function

$$\mathcal{W}_\epsilon(X, P, R, C) = f(X) + \frac{A}{2}\|P\|_F^2 + \frac{\mu B}{2}(\|R\|^2 + \|C\|^2) + \epsilon\langle X, P\rangle_F + \frac{\epsilon}{2}\|X\|_F^2, \tag{25}$$

where $A, B > 0$ and $\epsilon \in (0, \frac{1}{2})$ are constants to be chosen depending on h. For $\epsilon \le 1/2$ and $A, B \ge 1$, standard quadratic bounds give

$$\mathcal{W}_\epsilon \ge f(X) + \frac{1}{4}\|P\|_F^2 + \frac{\mu}{2}(\|R\|^2 + \|C\|^2) + \frac{\epsilon}{2}\|X\|_F^2, \tag{26}$$

$$\mathcal{W}_\epsilon \le f(X) + \frac{3}{4}\|P\|_F^2 + \frac{\mu}{2}(\|R\|^2 + \|C\|^2) + \frac{3\epsilon}{2}\|X\|_F^2. \tag{27}$$

We will show that for sufficiently small h and appropriate choices of A, B, ϵ, there exists $\rho > 0$ such that

$$\mathcal{W}_\epsilon(X_{n+1}, P_{n+1}, R_{n+1}, C_{n+1}) \le (1 - \rho h)\mathcal{W}_\epsilon(X_n, P_n, R_n, C_n).$$

The exponential convergence then follows by induction, using the boundedness of trajectories (which holds for small h by a discrete analogue of Lemma 2).

Bounding the effect of Step C′ and Step D. For brevity, let $\beta_n = \exp(-\tilde{\xi}_n h)$, with entries $\beta_{n,ij} \in (0, 1]$. Step C′ and Step D together give

$$\hat{P} = \mathrm{e}^{-\gamma h}\beta_n * P_n, \qquad \hat{R} = \mathrm{e}^{-\alpha h}\Big(R_n + \frac{1}{\mu}(1 - \beta_n^2) * ([P_n]^2\mathbf{1}_n)\Big),$$

and similarly for $\hat{C}$. Since $\beta_{n,ij} \le 1$, we have $\|\hat{P}\|_F \le \mathrm{e}^{-\gamma h}\|P_n\|_F$. For R, using $(1 - \beta_{n,ij}^2) \le 2h\xi_{n,ij} \le 2h\xi_{\max}$ (for small h), we obtain

$$\|\hat{R}\|^2 \le \mathrm{e}^{-2\alpha h}\Big(\|R_n\|^2 + \frac{Ch}{\mu}\|P_n\|_F^2\Big),$$

and similarly for $\|\hat{C}\|^2$, where $C > 0$ depends on $\xi_{\max}$.

Bounding the effect of Step B. Step B gives $P_{n+1} = \hat{P} - h\nabla f(X_n)$. Using the Lipschitz continuity of ∇f (from the Hessian bound M), we have

$$\begin{aligned}\|P_{n+1}\|_F^2 &= \|\hat{P}\|_F^2 - 2h\langle\hat{P}, \nabla f(X_n)\rangle_F + h^2\|\nabla f(X_n)\|_F^2 \\ &\le \mathrm{e}^{-2\gamma h}\|P_n\|_F^2 - 2h\mathrm{e}^{-\gamma h}\langle\beta_n * P_n, \nabla f(X_n)\rangle_F + M^2h^2\|X_n\|_F^2.\end{aligned}$$

Bounding the effect of Step A. $X_{n+1} = X_n + hP_{n+1}$, hence

$$\|X_{n+1}\|_F^2 = \|X_n\|_F^2 + 2h\langle X_n, P_{n+1}\rangle_F + h^2\|P_{n+1}\|_F^2.$$

Expanding the cross-term $\langle X_{n+1}, P_{n+1}\rangle_F$.

$$\begin{aligned}\langle X_{n+1}, P_{n+1}\rangle_F &= \langle X_n + hP_{n+1}, P_{n+1}\rangle_F \\ &= \langle X_n, \hat{P}\rangle_F - h\langle X_n, \nabla f(X_n)\rangle_F + h\|\hat{P}\|_F^2 + \mathcal{O}(h^2).\end{aligned}$$

The term $\langle X_n, \hat{P}\rangle_F$ can be written as $\mathrm{e}^{-\gamma h}\langle X_n, \beta_n * P_n\rangle_F$. Since $\beta_{n,ij} \in (0, 1]$, we have $\langle X_n, \beta_n * P_n\rangle_F \le \langle X_n, P_n\rangle_F + \|X_n\|_F\|P_n\|_F\|1 - \beta_n\|_\infty$, and $\|1 - \beta_n\|_\infty \le h\xi_{\max}$.

Assembling the Lyapunov difference. Using the above estimates and the smoothness of f,

$$f(X_{n+1}) \le f(X_n) + h\langle\nabla f(X_n), P_{n+1}\rangle_F + \frac{M}{2}h^2\|P_{n+1}\|_F^2.$$

Substitute $P_{n+1} = \hat{P} - h\nabla f(X_n)$:

$$f(X_{n+1}) \le f(X_n) + h\langle\nabla f(X_n), \hat{P}\rangle_F - h^2\|\nabla f(X_n)\|_F^2 + \frac{M}{2}h^2\|\hat{P}\|_F^2 + \mathcal{O}(h^3).$$

Now, plugging all estimates into $\mathcal{W}_\epsilon(X_{n+1}, P_{n+1}, R_{n+1}, C_{n+1})$ and using the strong convexity condition $\langle X_n, \nabla f(X_n)\rangle_F \geq af(X_n) + b\|X_n\|_F^2$, we obtain after careful bookkeeping (similar to the proof of Theorem 2 in Appendix H.3) an inequality of the form

$$\mathcal{W}_\epsilon(X_{n+1}, \dots) \leq \mathcal{W}_\epsilon(X_n, \dots) - h\Big(c_1 f(X_n) + c_2\|P_n\|_F^2 + c_3\|R_n\|^2 + c_4\|C_n\|^2 + c_5\|X_n\|_F^2\Big) + \mathcal{O}(h^2),$$

where the constants $c_i > 0$ can be ensured by choosing A, B appropriately (e.g., $A = 1$, B large enough to absorb the injection from the adaptive terms) and ϵ sufficiently small. The $\mathcal{O}(h^2)$ term is controlled by the boundedness of the trajectory, yielding a uniform bound $Kh^2\mathcal{W}_\epsilon$. Hence for $h < h^* = \min_i(c_i/K)$, we have

$$\mathcal{W}_\epsilon(X_{n+1}, \dots) \leq (1 - \rho h)\mathcal{W}_\epsilon(X_n, \dots),$$

with $\rho = \frac{1}{2}\min_i c_i$. Applying this recursively and using the lower bound (26) gives the claimed exponential convergence. $\square$